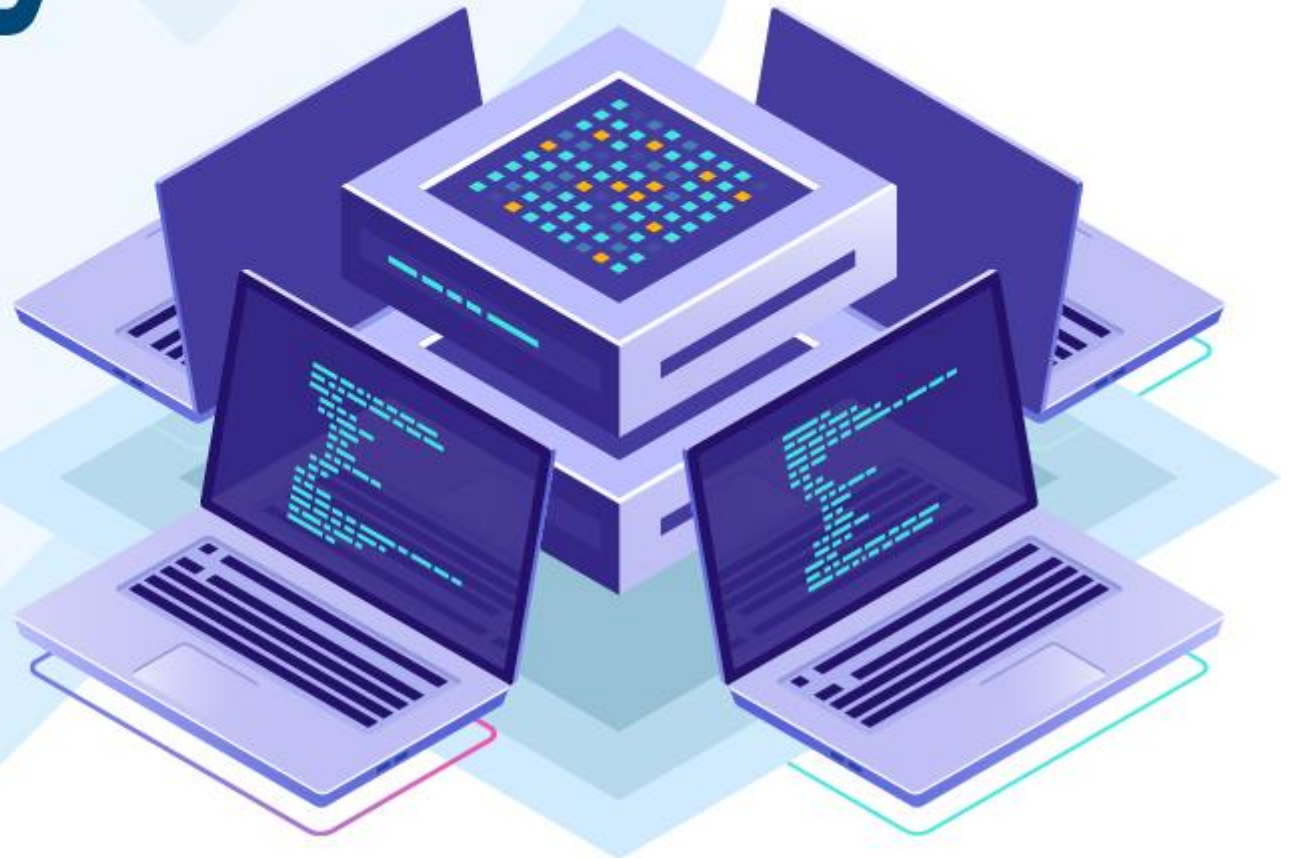


Information Technology

Spring Semester
2025 - 2026

» Semantic Web & Ontology
Lec. 1

» **Dr. Yasmine Mahmoud**



EELU

الجامعة المصرية للتعليم الإلكتروني
THE EGYPTIAN E-LEARNING UNIVERSITY

Semantic Web & Ontology

TEXTBOOK:

- 1. Foundations of Semantic Web Technologies** (Chapman & Hall/CRC Textbooks in Computing, by Pascal Hitzler, Markus Krötzsch, Sebastian Rudolph.

Student assessments + Assessment Grades

10 Project

10 Quizzes (2 quizzes each 5)

30 Midterm Exam

50 Final Exam

Section Attendance (minus)

Introduction to Semantic Web & Ontology

Introduction

- The **traditional internet** is mostly a **collection of websites connected by links**.
- When we **search for information** or browse, we're mostly looking at **how these pages are connected**.
- But **computers (machine)** don't fully understand the **meaning** of the information on these pages, they just **display** it.
- The **Semantic Web** is an idea to make the **internet smarter** by helping computers understand not just the words on a page, but what those words mean.

Traditional Web



- Most of **Traditional Web content** is suitable for human.
- Even when websites automatically create pages from databases (**like product listings or news articles**), they usually show the information in a visual format for humans (text, tables, images).
- They **do not include the original structured data** (like database fields, relationships, or categories) in a way that computers can easily understand.

Problems with traditional Web search

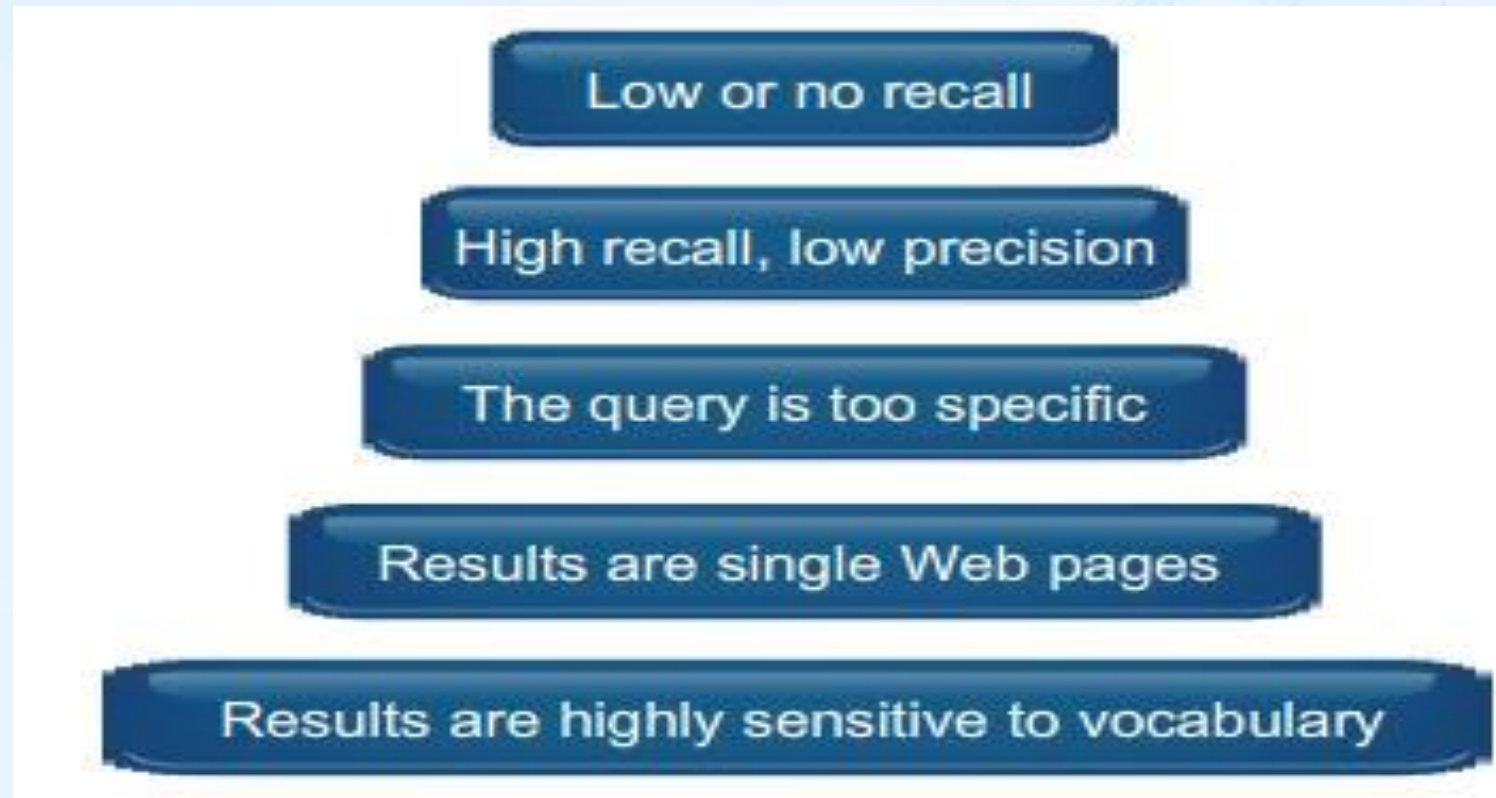


- Besides links that connect web pages to each other, the most useful tools are **search engines**.
- With the **huge mass of information** on the Web, it is almost impossible to look for material.
- **Keyword-based search engines** such as Google are the main tools for using Web.

Problems with traditional Web search



- However, there are **serious problems** associated with their use:



Problems with traditional Web search



- However, there are **serious problems** associated with their use:
 - **Recall** “is the number of pages retrieved as an answer to your query”.
 - **Precision** “is how many of the recalled sources are the ones you need”.

Problems with traditional Web search



1. Low or no recall:

- Often it happens that we don't get any relevant answer for our request, or that important and relevant pages are not retrieved.
- Although low recall is a less frequent problem with current search engines, it does occur.

Problems with traditional Web search



2. High recall, low precision:

- Even if the main relevant pages are retrieved, they are of **little** use if another 28,758 mildly relevant or irrelevant documents are also retrieved.
- Too much can easily become as bad as too little.

Problems with traditional Web search



3. The degree the query is specific:

- If the query is too specific; you may not retrieve any.
- If it is too broad, you will retrieve a lot of pages most of them are not relevant.

Problems with traditional Web search



4. Results are highly sensitive to vocabulary:

- Often the initial keywords do not get the results we want; in these cases, the relevant documents use different terminology from the original query.
- Simply, you have to use the exact word that exists in the document to have the search engine find it. If the same information is in a page but with a different wording, it will not be retrieved.
- This is **unsatisfactory** because semantically similar queries should return similar results.

Problems with traditional Web search



5. Results are single Web pages:

- If we need information that is spread over **various documents**, we must initiate several queries to collect the relevant documents, and then we must **manually** extract the partial information and put it together.

Problems with traditional Web search

- Even if a search is successful, it is the **person (human)** who must **browse** selected documents to **extract the information** he is looking for. That is, there is not much support for retrieving the information, a **very time-consuming** activity.
- Therefore, the term *information retrieval*, used in association with *search engines*, is somewhat misleading.
- *Location finder* might be a more appropriate term. Also, results of Web searches are not readily accessible by other software tools; search engines are often isolated applications.

1. What is the Semantic Web?

- The **Semantic Web** is an extension of the current web (World Wide Web) that aims to make information machine-readable and machine-understandable.
- Unlike the traditional web, where **data is primarily designed for humans to read** (e.g., HTML pages), the **Semantic Web** focuses on adding meaning (semantics) to data so that computers can interpret, process, and connect it intelligently.

1. What is the Semantic Web?

- **The Semantic Web:**

- **Definition:** The Semantic Web is a **vision** of the **web** where **data** is linked and enriched with **metadata**, enabling machines to **understand** and **reason** about the information.
- **Semantic web is used** to create a **web of data** that is interconnected and **meaningful**, allowing for **smarter search, automation, and integration of information across different sources.**

1. What is the Semantic Web?

- A world where your **computer** **doesn't just retrieve documents** for you but actually:
 - understands the content of those documents,
 - connects them to related information, and even
 - answers complex questions for you.
- That is the promise of the **Semantic Web**.

2. The Necessity of the Semantic Web: Its Importance

- The current web is a vast repository of information, but it has significant limitations:
 1. **Human-Centric:** Data is designed for humans to read, not for machines to process.
 - For example, a webpage about a **book** might display its **title**, **author**, and **summary** in a way that is **easy for a human to understand** but **difficult for a machine to extract and interpret**.

2. The Necessity of the Semantic Web: Its Importance

2. Lack of Context: Computers **cannot understand the meaning** of data.

- For instance, a search engine might return results for "Apple" without knowing whether you're looking for the **fruit**, the **tech company**, or the record label.

2. The Necessity of the Semantic Web: Its Importance.

3. Isolated Data: Information is often **isolated** in separate systems, making it **difficult to connect related data**.

- For example, a **hospital** might have patient records in one system, lab results in another, and prescriptions in a third, **with no way to link them automatically**.



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THANK YOU FOR WATCHING

QUESTIONS?

